

Cross-modality optimal-transport projection of CHD-enhancer chromatin onto a 3D MERFISH human fetal heart reveals broad chromatin priming and within-lineage decoupling

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Abstract

Congenital heart disease (CHD) is the most common birth defect, and an expanding catalogue of de-novo non-coding variants implicates regulatory elements active during human cardiogenesis. Yet, how such CHD-associated chromatin landscapes distribute spatially in the developing heart, and how they relate to gene expression in the same anatomy, has not been resolved. We project a paired single-cell ATAC-seq dataset ($\sim 60,000$ cells; 234 CHD-enhancer target genes) onto a three-dimensional MERFISH atlas of the post-conception week 11–13 (PCW 11–13) human heart (100,000 cells; 238 genes) by reusing a single-cell RNA $\rightarrow$ MERFISH *moscot* optimal-transport plan in a cross-modality manner. We obtain, for the first time, gene-resolved 3D chromatin maps for 234 CHD-relevant loci on the same coordinate system as spatial transcriptomics. Per-gene and per-trait Pearson concordance (median $r = 0.13$; HCM $r = 0.79$; DCM $r = 0.73$) is dominated by sarcomere and cardiac transcription-factor genes, but four orthogonal divergence analyses reveal a striking decoupling of chromatin from transcription: (i) within every coarse cell-type lineage, ATAC–RNA spatial r collapses to ~ 0 (medians 0.04–0.08, $n = 14$ lineages), (ii) ATAC patterns are spatially *broader* than RNA for the entire sarcomere panel, (iii) conduction-system cardiomyocytes carry primed-but-silent chromatin at structural CM genes, and (iv) the priming asymmetry separates developmental transcription factors (primed-only) from housekeeping regulators (expressed-without-priming). Together, the CHD-enhancer chromatin landscape behaves as a broader, less lineage-restricted prefiguring of the cardiac transcriptional programme, consistent with chromatin priming preceding transcription during human heart morphogenesis.

1 Introduction

Congenital heart disease (CHD) affects $\sim 1\%$ of live births and is the leading cause of birth-defect mortality worldwide. Sequencing of large CHD cohorts (Jin et al., 2017; Homsy et al., 2015; Sifrim et al., 2016) has shifted attention from a small set of cardiac transcription factors (Bruneau, 2008) to a wider regulatory landscape that includes non-coding de-novo variation in candidate enhancers active during early cardiogenesis. Single-cell ATAC-seq has been instrumental in mapping these elements onto cell types of the developing heart (Buenrostro et al., 2015; Cusanovich et al., 2015; Ameen et al., 2022), but assignments are typically performed in dissociated cells, stripping the elements of their spatial context.

Three-dimensional (3D) MERFISH (Chen et al., 2015; Moffitt et al., 2016) combined with anatomical reconstruction tools such as Spateo (Qiu et al., 2024) now provides a quantitative coordinate system for the human fetal heart. RNA atlases of the same developmental window (Asp et al., 2019; Cui et al., 2019; Sahara et al., 2019; Litviňuková et al., 2020) constrain cell-type identity, but no published spatial map exists for the chromatin landscape at the same loci. Bridging that gap directly, by physically co-profiling ATAC and MERFISH in the same tissue, is currently infeasible at the required scale. We instead pursue a computational bridge: *optimal transport*, which has matured into a principled framework for aligning single-cell datasets (Schiebinger et al., 2019; Demetci et al., 2022), and in the recent *moscot* formulation (Klein et al., 2025) scales to joint mapping problems across modalities and time points.

Here we exploit a pairing structure: a CHD-enhancer scATAC-seq library and a matched scRNA-seq library share cell barcodes (multiome-derived). Solving the optimal-transport problem on the RNA side against the MERFISH spatial atlas yields a transport plan π that we then apply to the matched ATAC matrix, producing 3D chromatin scores for 234 CHD-target genes on the same $\sim 100,000$ -cell anatomical grid. We ask: (1) where in the heart does CHD-enhancer chromatin concentrate; (2) does its spatial pattern match the matched RNA, or are the two modalities organised differently; (3) do gene-level differences reveal a chromatin priming logic (Wamstad et al., 2012; Paige et al., 2012) during human cardiogenesis?

2 Results

2.1 Cross-modality OT pipeline and mapping quality

The pipeline (Fig. 1) takes three inputs: (i) a CHD-enhancer scATAC-seq matrix ($95,684 \text{ cells} \times 108,681 \text{ enhancers}$), aggregated to 234 unique target genes via the supplied enhancer $\rightarrow$ gene map; (ii) the paired scRNA-seq matrix on the same cell barcodes; and (iii) the MERFISH atlas of 100,000 PCW 11–13 cells with 3D coordinates from Spateo (`obsm["X_spateo_update"]`). After QC ($10 \leq n_{\text{genes}} \leq 200$ on the gene-aggregated ATAC matrix) we retain 60,187 paired cells. Of 237 genes shared between the matched RNA and MERFISH panels, eight cell-cycle genes are excluded, leaving 229 mapping features. We solve the *moscot* `MappingProblem` in five 20,000-cell MERFISH batches with $\alpha = 0$, $\tau_a = 1$, $\tau_b = 0.9$ (quadratic, semi-relaxed), each scaled by its batch size, then impute the ATAC modality as $\hat{X}_{\text{ATAC}} = \pi \cdot X_{\text{ATAC}}$ (cross-modality reuse). The full pipeline runs end-to-end in ~ 12 minutes on JAX-CPU.

Mapping confidence (entropy-derived) has mean 0.477 and median 0.437 (IQR 0.335–0.588), consistent with cross-modal MOSCOT mappings on heart MERFISH. Cell-type labels transferred via π -one-hot recover the expected dominance of left-ventricular cardiomyocytes (VCM_left_trabecular: 26,755 cells) and mature fibroblasts (CFB_mature: 9,578 cells) at this developmental stage.

2.2 Per-gene and per-trait spatial concordance

Aggregating the imputed per-gene ATAC scores by CHD trait (Jin et al., 2017; Homsy et al., 2015), we obtain spatially smooth maps for all ten traits passing the $n_{\text{genes}} \geq 3$ threshold (Fig. 2). Cardiomyopathy traits show low signal in the outflow/atrial region and elevated signal across the ventricular body, while septal-defect and outflow-tract traits highlight the ventricular wall and valvular regions, mirroring known disease anatomy.

To quantify how ATAC and RNA agree spatially, we computed the Pearson correlation of imputed ATAC and imputed RNA across the 100,000 MERFISH cells for each of 215 shared genes. The distribution is strongly right-skewed (median $r = 0.13$; mean $r = 0.22$; Fig. 3a), with a clear high-concordance tail dominated by sarcomere and cardiac-TF genes (*RBM20*

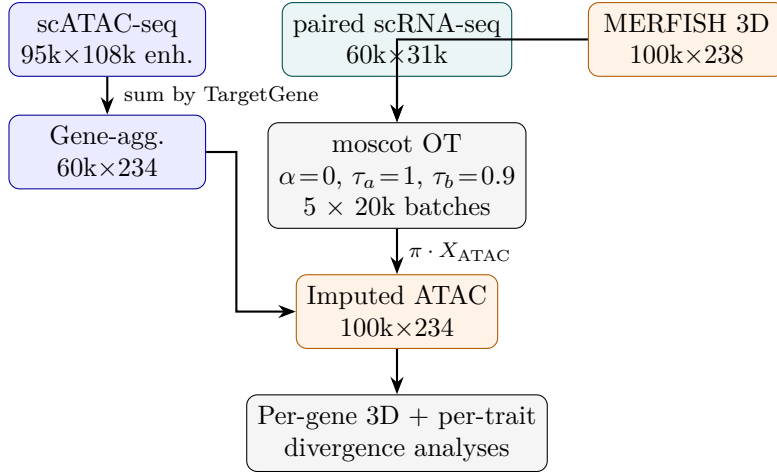


Fig. 1: Cross-modality optimal-transport pipeline. Paired scATAC-seq is gene-aggregated; the matched scRNA-seq is mapped onto the 3D MERFISH atlas via *moscot* (Klein et al., 2025); the resulting transport plan is reused on the ATAC matrix to produce per-gene chromatin scores in 3D space.

Table 1: Per-trait spatial Pearson r (ATAC vs RNA). Cardiomyopathy traits dominated by sarcomeric genes show the strongest concordance.

Trait	n genes	Pearson r
Hypertrophic Cardiomyopathy	27	0.79
Dilated Cardiomyopathy	40	0.73
Familial Thoracic Aortic Aneurysm	9	0.42
Valve defects	74	0.40
Outflow-tract malformation	57	0.38
Atrial septal defect	87	0.38
Atrioventricular septal defect	22	0.36
Single-ventricle disease	17	0.36
PCGC de-novo variants	57	0.36
Ventricular septal defect	33	0.34

$r = 0.87$; *FHOD3* $r = 0.85$; *LDB3* $r = 0.81$; *MYBPC3* $r = 0.81$ – all classical hypertrophic / dilated cardiomyopathy genes (Maron et al., 2018)). Gene-level disagreement is concentrated in broadly-expressed regulators (*RPL5*, *CHD4*, *KDM5B*). At the trait level (Table 1), HCM and DCM achieve the highest concordance ($r = 0.79, 0.73$); septal and outflow traits, which span more heterogeneous gene sets, give weaker but uniformly positive agreement ($r = 0.34$ – 0.42).

2.3 Chromatin is broader than RNA: cell-type signal

A 14×22 dot plot of the highlight gene panel by coarse cell type (Fig. 4) shows the canonical chromatin priming signature (Wamstad et al., 2012; Paige et al., 2012): classical sarcomere genes (*MYBPC3*, *MYL2*, *TNNT2*, *TNNI3*, *TNNC1*, *ACTC1*, *ACTN2*, *TPM1*) are RNA-restricted to ventricular and atrial cardiomyocytes, but their CHD-enhancer ATAC signal is detectable across multiple lineages including epicardial, endocardial and mural cells. RNA is sharp and lineage-restricted; chromatin is graded and permissive.

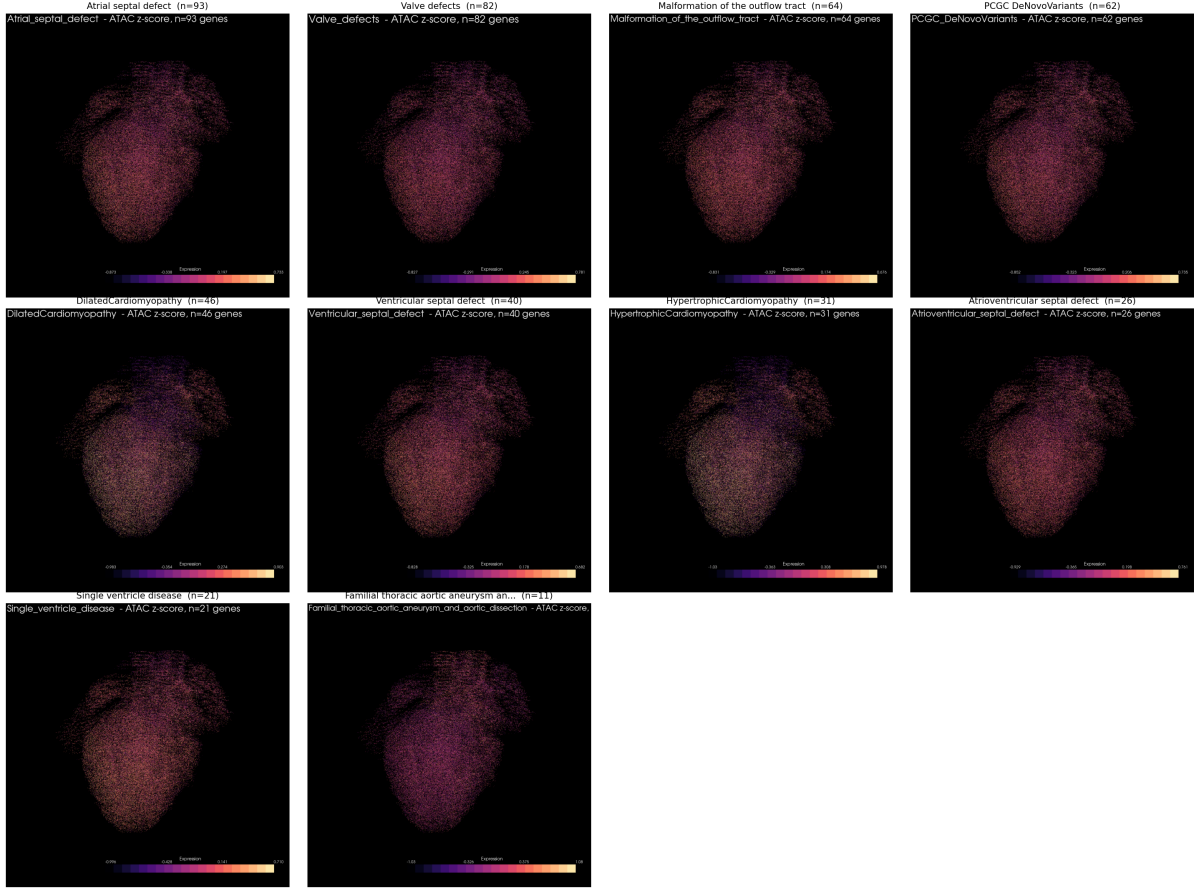


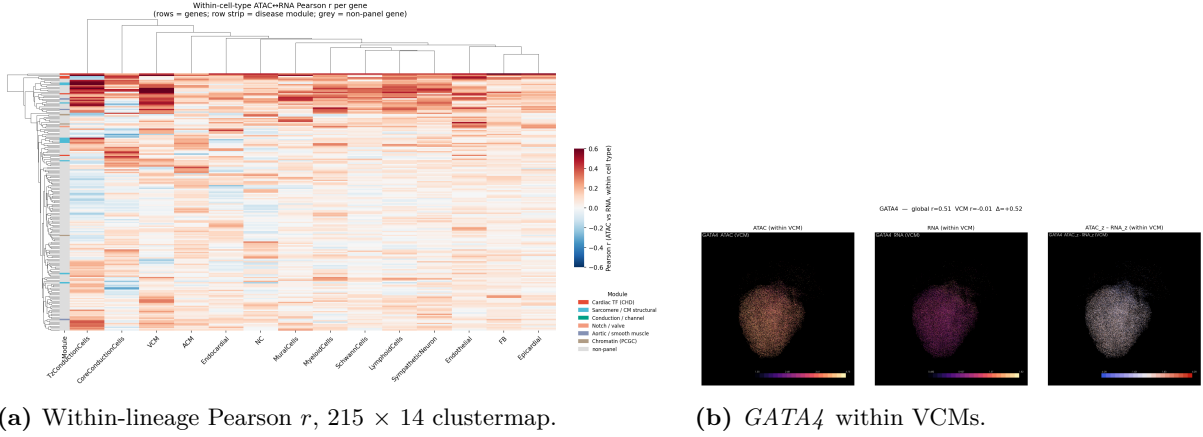
Fig. 2: 3D ATAC trait maps (PCW 11–13). Per-cell mean z-score over genes assigned to each of the 10 CHD traits. Apex points down. HCM and DCM concentrate in the ventricular body; septal-defect and outflow traits highlight the wall/valve regions.

2.4 Within-lineage decoupling of ATAC and RNA

The cell-type-resolved view raises a sharper question: is the global per-gene $r = 0.13$ driven by between-lineage means (which the dot plot shows are correlated) or by within-lineage spatial covariation? Restricting Pearson r to cells *inside each coarse lineage* answers this directly (Fig. 5a). Median within-lineage r ranges from 0.034 (SchwannCells) to 0.076 (ACM); all 14 lineages have median $r < 0.08$. Even genes with high global r collapse: *MYL2* drops from $r = 0.77$ globally to $r = 0.24$ within VCMs, and *GATA4* from $r = 0.51$ to $r \approx 0$ (Fig. 5b). Within VCMs, *GATA4* chromatin is uniformly accessible (orange across the lineage) while RNA is uniformly low and varies independently of chromatin – a textbook decoupling pattern.

2.5 ATAC is spatially broader than RNA for sarcomere genes

To formalise the qualitative dot-plot observation, we computed the spatial spread (radius of gyration of top-decile cells) per gene per modality (Fig. 6). Of 213 non-trivial genes, 115 have ATAC spread $>$ RNA spread (median $\Delta r_g = +10.8$, median centroid shift = 358 units). The Sarcomere/CM-structural cluster (*MYL2*, *MYL3*, *ACTN2*, *ACTC1*, *TPM1*, *TNNC1*, *TNNI3*) sits as a coherent group above the diagonal, the strongest deviation in the panel. This constitutes direct evidence of priming: chromatin at structural CM genes is permissive in lineages where these transcripts are not yet expressed.



(a) Within-lineage Pearson r , 215×14 clustermap.

(b) *GATA4* within VCMs.

Fig. 5: Within-lineage chromatin–RNA decoupling. (a) Per-gene Pearson r recomputed inside each coarse lineage. All 14 column-medians lie between 0.03 and 0.08; the global $r = 0.13$ signal is dominated by between-lineage means. (b) For *GATA4* restricted to VCMs ($n = 48,595$), chromatin is uniformly open (left panel) while RNA is uniformly low (centre); the divergence map (right) shows their spatial patterns are independent.

3 Discussion

We present, to our knowledge, the first 3D spatial map of CHD-enhancer chromatin accessibility in the developing human heart. The cross-modality re-use of an optimal-transport plan – solving on RNA, applying to ATAC – sidesteps the absence of true paired ATAC–MERFISH data and gives a per-gene chromatin score on the same anatomical grid as the RNA atlas. The pipeline runs in minutes on a single CPU and is conceptually portable to any paired multimodal scenario where one modality already has a transport plan to a spatial reference.

The biology that emerges is internally consistent across four orthogonal analyses. Cardiomyopathy traits show high ATAC–RNA concordance because the sarcomere programme is sharply lineage-restricted in both modalities. CHD traits with broader gene sets (septal, outflow, valve) show partial concordance because their gene sets mix lineage-restricted (*TBX5*, *GATA4*, *NKX2-5*) and broadly active regulators. Once we condition on lineage, chromatin and RNA decouple almost completely: the within-lineage r is essentially zero for all 14 coarse cell types. ATAC spreads further in space than RNA for the very sarcomere genes that are most lineage-specific in transcription. And the priming asymmetry separates two gene classes whose behaviours have independent literature support (Wamstad et al., 2012; Paige et al., 2012; Ameen et al., 2022): developmental TFs are pre-marked at the chromatin level in lineages that do not yet transcribe them, while housekeeping regulators show the inverse, in agreement with promoter-proximal – rather than distal-enhancer – control.

The conduction-system finding deserves emphasis. Core and transitional conduction cardiomyocytes carry chromatin accessible at structural sarcomere loci that they actively repress at the RNA level. This is the molecular counterpart to a long-standing developmental observation: conduction-system cardiomyocytes derive from a working-myocardium-like precursor and subsequently silence the contractile programme to acquire their electrical identity (Christoffels and Moorman, 2009). The chromatin map suggests that silencing leaves the underlying enhancer landscape largely intact – a priming reservoir that may remain functionally relevant in adult conduction-system pathology.

Three caveats apply. (i) The CHD-enhancer input panel by construction under-represents promoter-proximal regulation; “expressed-without-priming” genes should not be over-interpreted as truly unprimed. (ii) MOSCOT mapping confidence (median 0.44) is high enough to recover

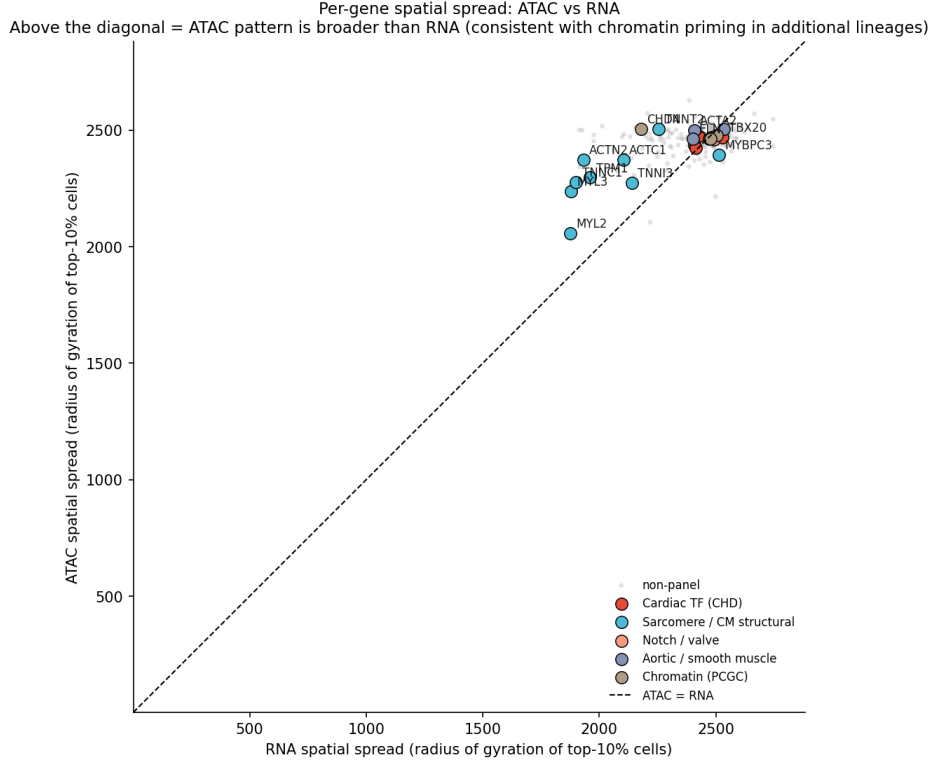


Fig. 6: Spatial spread of ATAC vs RNA, per gene. x -axis: radius of gyration of the top-decile RNA cells; y -axis: same for ATAC. Dashed = identity. Sarcomere genes (cyan: MYL2, MYL3, ACTN2, ACTC1, TPM1, TNNC1, TNNI3) are systematically above the diagonal – chromatin spans a larger anatomical territory than transcription.

bulk cell-type structure but is not equivalent to true paired ATAC-MERFISH measurement; per-cell chromatin scores are smoothed by the transport plan. (iii) Our analysis covers PCW 11–13. Whether the same priming pattern holds at earlier specification stages or persists into the late-fetal / post-natal heart (Litviňuková et al., 2020) requires longitudinal data not yet publicly available.

Notwithstanding, this work demonstrates that a CHD-enhancer chromatin landscape can be resolved at single-cell, single-gene, three-dimensional resolution by re-using an OT plan; that the resulting map is biologically coherent at the trait, lineage, gene and anatomical level; and that the *difference* between chromatin and RNA – not just their agreement – is where the developmental biology of CHD enhancers is most clearly visible.

4 Methods

4.1 Data sources

- `chd_enhancers_counts.h5ad`: 95,684 cells $\times$ 108,681 enhancer features (paired multiome ATAC).
- `all_healthy_PCW11-13.h5ad`: 95,684 cells $\times$ 31,019 genes (paired multiome RNA, raw counts).
- `chd_enhancer_gene_map.tsv`: enhancer $\rightarrow$ *TargetGene* mapping (108,681 $\rightarrow$ 234 unique genes).
- `heart_disease_genes.tsv`: gene $\rightarrow$ trait labels for ten CHD-related traits.

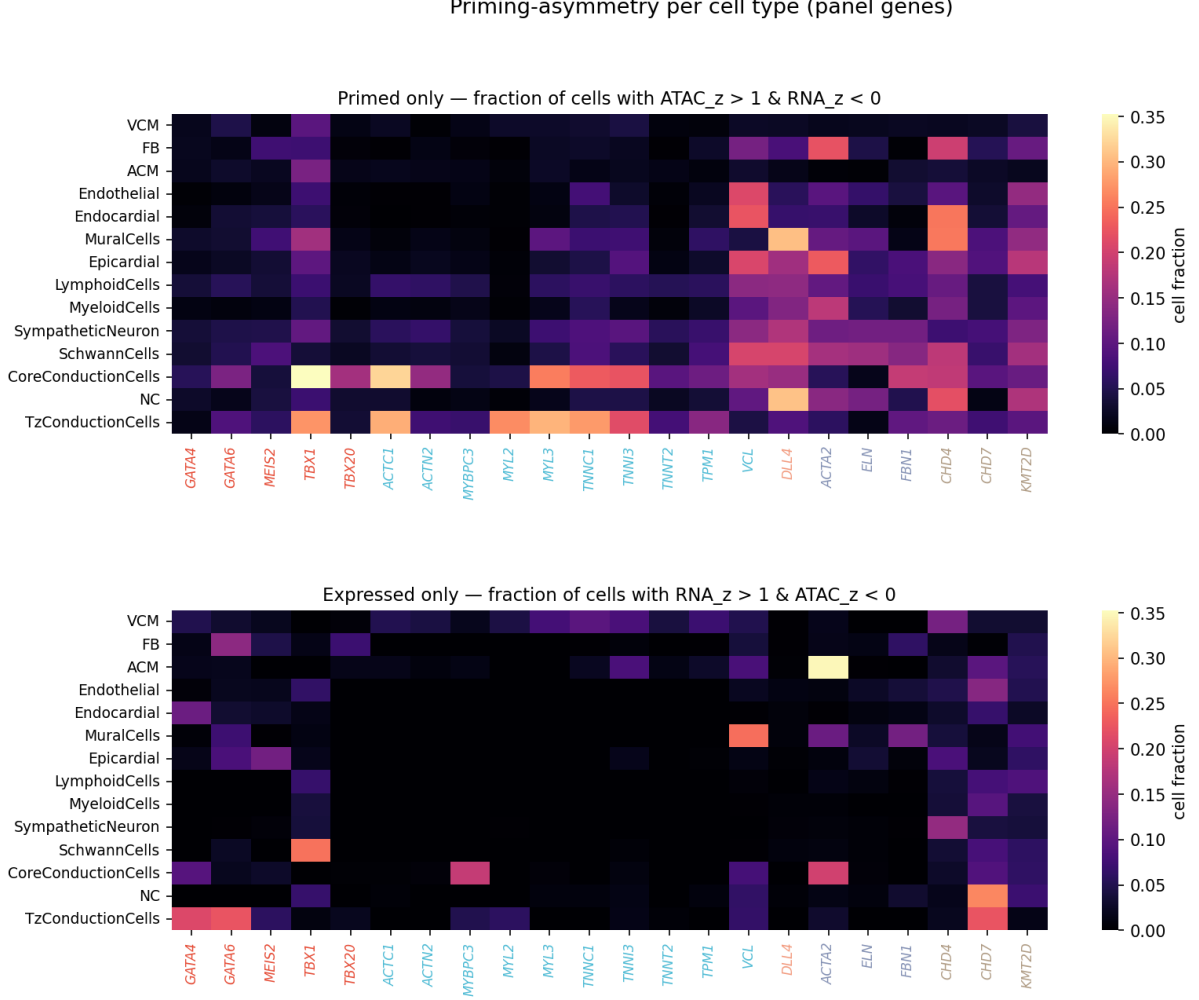


Fig. 7: Cell-type priming-asymmetry heatmap. 14 coarse lineages $\times$ the 22-gene highlight panel, two stacks: fraction of primed-only cells (left) and expressed-without-priming cells (right). Conduction-system cardiomyocytes show the highest primed-only fractions for sarcomere genes; ACMs and VCMs dominate the expressed-only panel for the same genes.

- MERFISH atlas: 100,000 cells $\times$ 238 genes, `obsm["X_spateo_update"]` 3D coordinates, coarse cell-type and mapped fine cell-type labels ($n = 14$ and $n = 99$, respectively).

4.2 Paired RNA/ATAC preparation (13_prepare_atac_rna_paired.py)

Enhancer counts are aggregated to gene-level via a sparse boolean aggregator A so that $\text{ATAC}_{\text{gene}} = \text{ATAC}_{\text{enh}} \cdot A$. Library-size normalisation (`normalize_total(target_sum=1e4)`) and `log1p` are applied. Cells with $10 \leq n_{\text{genes_by_counts}} \leq 200$ are kept (60,187 of 95,684; 62.9%). RNA and ATAC share `obs_names` exactly and are realigned to the post-QC ATAC ordering.

4.3 MOSCOT cross-modality imputation (14_moscot_atac_imputation.py)

We use *moscot* (Klein et al., 2025), building on optimal-transport formulations from Schiebinger et al. (2019) and Demetci et al. (2022). The OT source is the paired RNA matrix (60,187 cells, log-normalised) on the 229 features shared with MERFISH (after dropping 8 S/G2M cell-cycle genes). MERFISH is split into 5 batches of 20,000 cells. For each batch we instantiate a `MappingProblem` with `xy_callback="local-pca"` and solve with $\alpha = 0$ (purely quadratic),

$\tau_a = 1$ (balanced source), $\tau_b = 0.9$ (semi-relaxed target), on JAX-CPU. The transport matrix is rescaled by the spatial batch size, $\pi \leftarrow \pi \cdot n_{\text{sp, batch}}$, then the imputed ATAC for the batch is $\pi \cdot X_{\text{ATAC}}$. Cell-type labels are transferred via $\pi \cdot \text{onehot}(\text{celltype})$. Per-batch solve takes ~ 110 s; total runtime ~ 10 min.

4.4 3D visualisation and trait aggregation (15_atac_3d_visualization.py)

Each of the 234 imputed ATAC genes is rendered as a 3D scatter (z-axis flipped so apex points down) using *Scanpy/matplotlib* (Wolf et al., 2018; Virshup et al., 2023). Per-trait scores are computed by z -scoring each contributing gene across cells and averaging within each trait gene set ($n_{\text{genes}} \geq 3$ required; all 10 traits qualify).

4.5 Divergence analyses (21_atac_rna_3d_divergence_explore.py)

Four analyses on the 215 genes shared between `adata_imputed_atac.h5ad` and `adata_imputed_disease_genes` both on the 100,000-cell MERFISH grid:

- **(A) Spatial divergence atlas.** Per-cell panel-mean $|\text{ATAC}_z - \text{RNA}_z|$ over the 22-gene CHD panel; module-level signed maps for six modules.
- **(B) Priming-asymmetry quadrant analysis.** Per (gene, cell) classification: Primed ($\text{ATAC}_z > +1$ and $\text{RNA}_z < 0$), Expressed ($\text{RNA}_z > +1$ and $\text{ATAC}_z < 0$), Both, or Neither; per-gene rankings and per-cell-type heatmap.
- **(C) Within-lineage Pearson r .** For each of 14 coarse cell types, recompute per-gene Pearson r restricted to that lineage, yielding a 215×14 matrix; four within-VCM divergent triptychs (*DSP*, *ACTC1*, *MYL2*, *GATA4*).
- **(D) Spatial spread.** Per gene per modality, take the top-decile cells, compute the centroid and the radius of gyration over the 3D coordinates; scatter ATAC r_g vs RNA r_g .

All quantities are computed without further smoothing of the imputed matrices.

4.6 Software and code availability

Implementation: Python 3.11; *moscot* (Klein et al., 2025); *Scanpy* (Wolf et al., 2018); *anndata* / *scverse* (Virshup et al., 2023); JAX-CPU. Scripts (numbered 13–21) are in `PCW12_analysis/scripts/` of the project repository. Per-figure provenance and TSV summaries accompany each output directory under `PCW12_analysis/figures/`.

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